Series &RQPS

SET-5

प्रश्न-पत्र कोड
Q.P. Code **56(B)**

रोल नं.
Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

**नोट**

(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं ।

(II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में **33** प्रश्न हैं ।

(III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।

(IV) **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**

(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।

**NOTE**

(I) Please check that this question paper contains **23** printed pages.

(II) Please check that this question paper contains **33** questions.

(III) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(IV) **Please write down the serial number of the question in the answer-book before attempting it.**

(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

# रसायन विज्ञान (सैद्धांतिक)

## (केवल दृष्टिबाधित परीक्षार्थियों के लिए)

# CHEMISTRY (Theory)

## (FOR VISUALLY IMPAIRED CANDIDATES ONLY)

*निर्धारित समय : 3 घण्टे* | *अधिकतम अंक : 70*

*Time allowed : 3 hours* | *Maximum Marks : 70*

***General Instructions :***

*Read the following instructions carefully and follow them :*

*(i)* *This question paper contains **33** questions. **All** questions are **compulsory**.*

*(ii)* *This question paper is divided into **five** sections – **Section A**, **B, C, D** and **E**.*

*(iii)* ***Section A** – questions number **1** to **16** are multiple choice type questions. Each question carries **1** mark.*

*(iv)* ***Section B** – questions number **17** to **21** are very short answer type questions. Each question carries **2** marks.*

*(v)* ***Section C** – questions number **22** to **28** are short answer type questions. Each question carries **3** marks.*

*(vi)* ***Section D** – questions number **29** and **30** are case-based questions. Each question carries **4** marks.*

*(vii)* ***Section E** – questions number **31** to **33** are long answer type questions. Each question carries **5** marks.*

*(viii)* *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the sections except Section A.*

*(ix)* *Use of calculators is **not** allowed.*

## SECTION A

*Questions no. **1** to **16** are Multiple Choice type Questions, carrying **1** mark each.* *16×1=16*

**1.** Low concentration of oxygen in the blood and tissues of people living at high altitudes is due to :

(A) Low temperature

(B) Low atmospheric pressure

(C) High atmospheric pressure

(D) High temperature

**2.** The quantity of charge required to obtain one mole of Al from $Al_2O_3$ is :

(A) 1 F (B) 6 F

(C) 3 F (D) 2 F

**3.** Which of the following oxidation state is common for all lanthanoids ?

(A) + 2 (B) + 4

(C) + 6 (D) + 3

**4.** Which of the following ions has the electronic configuration $3d^3$ ? [Atomic number : Cr = 24, Mn = 25, Fe = 26, Co = 27]

(A) $Cr^{3+}$ (B) $Mn^{3+}$

(C) $Fe^{3+}$ (D) $Co^{3+}$

**5.** When 1 mole of $CoCl_3 . 5NH_3$ is treated with excess of $AgNO_3$, 2 moles of AgCl are obtained. The secondary valency of Co will be :

(A) 6 (B) 4

(C) 3 (D) 5

**6.** A primary alkyl halide would prefer to undergo ___________.

(A) $S_N2$ reaction

(B) $S_N1$ reaction

(C) Electrophilic substitution reaction

(D) Racemisation

**7.** Phenol is less acidic than :

(A) Ethanol

(B) *o*-cresol

(C) *p*-cresol

(D) *o*-nitrophenol

**8.** Reduction of aromatic nitro compounds using Sn and HCl gives :

(A) Aromatic amide

(B) Aromatic oxime

(C) Aromatic primary amine

(D) Aromatic hydrocarbon

**9.** Nucleic acids are the polymers of :

(A) Nucleotides

(B) Nucleosides

(C) Bases

(D) Sugars

**10.** Methylamine reacts with $HNO_2$ to form :

(A) $CH_3 - O - N = O$

(B) $CH_3 - O - CH_3$

(C) $CH_3 - OH$

(D) $CH_3 - CHO$

**11.** Which one of the following has the lowest $pK_b$ value ?

(A) Ammonia

(B) Aniline

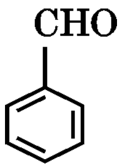
(C) Methylamine

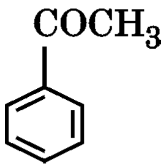
(D) Dimethylamine

**12.** Which of the following compounds is most reactive towards nucleophilic addition reactions ?

(A) $CH_3CHO$

(B) $CH_3 - CO - CH_3$

(C) 

(D) 

***For Questions number 13 to 16, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.***

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

**13.** *Assertion (A) :* Vitamin C cannot be stored in our body.

*Reason (R) :* Vitamin C is fat soluble and is excreted from the body in urine.

**14.** *Assertion (A) :* Bromination of benzoic acid gives m-bromobenzoic acid.

*Reason (R) :* Carboxyl group acts as a deactivating and meta-directing group.

15. *Assertion (A) :* When NaCl is added to water, a depression in freezing point is observed.

*Reason (R) :* The lowering of vapour pressure of a solution causes depression in the freezing point.

16. *Assertion (A) :* Linkage isomerism arises in coordination compounds containing ambidentate ligand.

*Reason (R) :* Ambidentate ligand has two same donor atoms.

## SECTION B

17. Why does a solution containing non-volatile solute have higher boiling point than a pure solvent ? Why is elevation of boiling point a colligative property ? *2*

18. (a) Write the formulae of the following compounds : *2×1=2*

(i) Potassium tetrahydroxidozincate (II)

(ii) Hexaammineplatinum (IV) chloride

**OR**

(b) Define the following terms : *2×1=2*

(i) Homoleptic complex

(ii) Chelate ligand

19. (a) Out of $(CH_3)_3C - Br$ and $(CH_3)_3C - I$, which one is more reactive towards $S_N1$ and why ?

(b) Write the product formed when p-nitrochlorobenzene is heated with aqueous NaOH at 443 K followed by acidification. *2×1=2*

**20.** Write the reactions involved in the following : *2×1=2*

(a) Reimer-Tiemann reaction

(b) Friedel-Crafts alkylation of anisole

**21.** A and B are two functional isomers of compound $C_3H_6O$. On heating with NaOH and $I_2$, isomer B forms a precipitate of iodoform whereas isomer A does not form any precipitate. Write the formulae of A and B. *2*

## SECTION C

**22.** Calculate the boiling point of the solution when 4 g of $MgSO_4$ ($M = 120\ g\ mol^{-1}$) was dissolved in 100 g of water, assuming $MgSO_4$ undergoes complete ionization. *3*

($K_b$ for water = $0{\cdot}52\ K\ kg\ mol^{-1}$)

**23.** Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration. *3*

**24.** A reaction is first order in A and second order in B. *3×1=3*

(a) Write the differential rate equation.

(b) How is the rate affected on increasing the concentration of B three times ?

(c) How is the rate affected when the concentrations of both A and B are doubled ?

**25.** Define Crystal field splitting energy. On the basis of Crystal field theory, write the electronic configuration of $d^5$ ion when : *3*

(a) $\Delta_o > P$

(b) $\Delta_o < P$

**26.** Write the main products when : (any ***three***) *3×1=3*

(a) Methyl chloride is treated with KCN.

(b) 2,4,6-trinitrochlorobenzene is subjected to hydrolysis.

(c) Methyl chloride is treated with sodium in the presence of dry ether.

(d) n-butylchloride is treated with alcoholic KOH.

**27.** Explain the following : *3×1=3*

(a) Carbylamine reaction

(b) Hoffmann bromamide degradation reaction

(c) Ammonolysis

**28.** How do you convert the following ? *3×1=3*

(a) Propan-2-ol to Propanone

(b) Phenol to Benzene

(c) Propene to Propan-2-ol

## SECTION D

*The following questions are case-based questions. Read the case carefully and answer the questions that follow.*

**29.** Rate of reaction depends upon the experimental conditions such as concentration of reactants, temperature and catalyst. Mathematical representation of rate of a reaction is given by rate law :

$$\text{Rate} = k\,[A]^p\,[B]^q$$

A balanced chemical equation never gives us a true picture of how a reaction takes place since rarely does a reaction get completed in one step. The reactions taking place in one step are called elementary reactions. When a sequence of elementary reactions give us the products, the reactions are called complex reactions. Order and molecularity of an elementary reaction are same.

(a) For a reaction A + B $\rightarrow$ Product, the rate law is given by

$$\text{Rate} = k\,[A]^1\,[B]^{1/2}$$

What is the overall order of the reaction ? *1*

(b) How are order and molecularity different for complex reactions ? *1*

(c) For a first order reaction, show that the time required for 99% completion is twice the time required for the completion of 90% of reaction. [Given : log 10 = 1] *2*

**OR**

(c) What is meant by zero order reaction ? Give any one example of zero order reaction. *2*

**30.** Aldehydes and ketones undergo nucleophilic addition reactions onto the carbonyl group but carboxylic acids do not undergo nucleophilic addition reaction. The alpha ($\alpha$)-hydrogens of aldehydes and ketones are acidic in nature. Therefore aldehydes and ketones having at least one $\alpha$-hydrogen undergo Aldol condensation. Aldehydes are easily oxidized by mild oxidizing agents such as Tollens' reagent and Fehling's reagent.

(a) Why does methanal not undergo Aldol condensation ? *1*

(b) Why are $\alpha$-hydrogens of aldehydes and ketones acidic in nature ? *1*

(c) (i) Why are aldehydes easily oxidised as compared to ketones ?

(ii) Why is addition of sodium hydrogen sulphite to aldehydes used for separation and purification of aldehydes ? *2*

**OR**

(c) What happens when :

(i) ethanal is treated with dilute NaOH ?

(ii) propanone is treated with HCN ? *2*

## SECTION E

**31.** Do any ***five*** of the following : *5×1=5*

(a) Pentaacetate of glucose does not react with hydroxylamine. Give reason.

(b) Why do amino acids behave like salts ?

(c) Why must water soluble vitamins be taken regularly in diet ?

(d) Why are the two strands in DNA complementary to each other ?

(e) What happens when D-glucose is heated with HI ?

(f) What is meant by glyosidic linkage ?

(g) What are enzymes ?

**32.** The elements of 3d transition series are given as :

Sc Ti V Cr Mn Fe Co Ni Cu Zn

(a) Answer the following : *5×1=5*

(i) Which element shows +1 oxidation state ?

(ii) Which element is a strong reducing agent in +2 oxidation state and why ?

(iii) Write the element which shows maximum number of oxidation states. Give reason.

(iv) Which element has the lowest value of enthalpy of atomization and why ?

(v) Which element shows only +3 oxidation state ?

**OR**

(b) (i) Account for the following :

(1) $Cu^{2+}$ salts are coloured while $Zn^{2+}$ salts are white.

(2) $Mn^{3+}$ is a strong oxidising agent.

(3) Transition metals and their many compounds act as good catalysts.

(ii) What is lanthanoid contraction ? Write one consequence of lanthanoid contraction. *3+2=5*

**33.** (a) (i) Depict the galvanic cell in which the reaction

$$Zn\ (s) + 2Ag^{+}\ (aq) \rightarrow Zn^{2+}\ (aq) + 2Ag\ (s)$$

takes place. Further show :

(1) which of the electrodes is negatively charged ?

(2) the carriers of the current in the cell.

(ii) Define fuel cell. Write two fuels that can be used in fuel cells. *3+2=5*

**OR**

(b) (i) State the following :

(1) Kohlrausch law

(2) Faraday's first law of electrolysis

(ii) Define the following :

(1) Corrosion

(2) Primary batteries and secondary batteries

Give one example in each case. *2+3=5*